

Nakuja Project

Solid Propulsion Team

One-Grain Static Test — 1% Sodium Bicarbonate Suppressant Trial

Document Version: 1.0 (Post-Execution Report)
Time of Test: Tuesday, 21st July 2026
Motor Configuration: N4, 1-Grain, Aluminium Casing, BATES Geometry
Propellant Additive: 1% Sodium Bicarbonate (NaHCO_3) suppressant
Prepared by: Paschal Peters (Data Recorder / Timer)
Outcome: **SUCCESS** (Combustion) — Good burn, minor nozzle flare noted
Instrumentation: **FAILED** — Load cell malfunction, no thrust curve captured

1 Purpose

This report documents a static test conducted on a grain prepared with **1% Sodium Bicarbonate (NaHCO_3)** added as a combustion suppressant, continuing the team's investigation into propellant additives following the surfactant trials documented previously. Unlike the 0.1% SLS surfactant test, which produced severe, unwanted combustion suppression, this test is classified as a combustion success: the motor produced a good, sustained burn with a reasonable duration and a healthy exhaust stream, with only a minor cosmetic observation — brief flaring at the nozzle toward the end of the burn — noted for the record. However, the **load cell malfunctioned during the test**, and no thrust curve was captured. This report therefore documents a successful burn with an instrumentation gap, rather than a fully characterised test.

2 Test Overview

Motor	N4, 1-Grain, Aluminium Casing
Additive	1% Sodium Bicarbonate (suppressant)
Burn Time	≈ 12 s
Exhaust Stream Quality	Good
Anomalies (Combustion)	Brief flaring at the nozzle toward the end of the burn
Instrumentation	Load cell malfunction — no thrust curve captured
Outcome	Success (combustion); Failed (thrust data)

3 Observations

3.1 Burn Quality and Duration

The motor produced a **good, sustained burn** with a duration of approximately **12 seconds**, described as fairly good relative to expectations for this configuration. This stands in clear contrast to the earlier 0.1% SLS surfactant test, where burn time extended to several minutes with negligible thrust — here, the suppressant produced a burn that remained controlled and complete without the extreme rate suppression seen in that prior trial.

3.2 Exhaust Stream

The stream of combustion gases exiting the nozzle was reported as good throughout the majority of the burn, consistent with steady, well-behaved combustion.

3.3 Nozzle Flaring (Late-Burn)

Toward the end of the burn, a brief instance of visible fire/flaring was observed at the nozzle. This was noted as a minor observation rather than a fault — no other combustion issues were reported.

3.4 Load Cell Malfunction — No Thrust Data

The load cell **malfunctioned during the test**, and as a result **no thrust curve was captured** for this burn. This is an instrumentation failure independent of the combustion outcome — the motor itself performed well, but no quantitative thrust, peak value, or impulse data is available to characterise this burn numerically. This is the second consecutive test (after the 0.1% SLS trial) without usable thrust data, though for different underlying reasons: the SLS test produced insufficient thrust for the load cell to register, whereas here the load cell itself failed despite the burn being of a magnitude that should have been well within its measurable range.

4 Discussion

The 1% sodium bicarbonate suppressant appears to have moderated the burn in a controlled, useful way — a fairly good ~ 12 s burn duration with a good exhaust stream — without the runaway rate suppression seen when the SLS surfactant dose was set too high in the previous trial. This suggests sodium bicarbonate at this concentration is a promising direction for controlling burn characteristics without compromising overall combustion quality.

The late-burn nozzle flaring is worth noting for future tests, as it is a common indicator of tail-off effects such as incomplete consumption of the grain's outer surface, minor unburned residue reaching the nozzle late in the burn, or a brief afterburning of gases as chamber pressure drops toward the end of the burn. Since no other issues were observed and the overall test was successful, this is recorded here primarily as an observation to track across future tests with this suppressant, rather than as a fault requiring immediate corrective action.

5 Outcome Classification

This test is classified as a **combustion SUCCESS** with a **data-capture FAIL**. The motor achieved a good, controlled ~ 12 s burn with a healthy exhaust stream, and the only combustion-related anomaly — brief nozzle flaring near the end of the burn — did not affect the overall outcome or indicate a fault requiring corrective action. However, the load cell malfunction means this test cannot contribute quantitative thrust, peak, or impulse data to the team's records, which limits its value for comparison against other additive trials.

6 Recommendations and Next Steps

- **Inspect and repair/recalibrate the load cell** before the next static test — this is now the second test in a row without usable thrust data, and repeated instrumentation gaps will prevent proper comparison across the additive trials underway.
- Continue evaluating 1% sodium bicarbonate as a viable suppressant concentration, given the good burn quality and duration achieved in this test, and prioritise re-testing it with working instrumentation to obtain the missing thrust curve.
- Track the late-burn nozzle flaring observation across future tests with this suppressant to determine whether it is a repeatable characteristic of this formulation or an isolated occurrence.
- Where available, review video footage for the final seconds of the burn to characterise the flaring event more precisely (duration, intensity) for comparison against future runs, since thrust data is unavailable for this test.
- Consider this test as a positive comparison point against the 0.1% SLS surfactant trial when evaluating which additives are suitable for controlled burn-rate moderation versus which risk excessive suppression — noting that neither test currently has a captured thrust curve, for different reasons.